COMSATS University Islamabad (Lahore Campus)
Department of Computer Engineering

Member B Sprint Plan

Edge / CV Development Lead

5 Months · 5 Sprints — Your Tasks Only

Master timeline: `.sprint/centralized_sprint_plan.tex`

Spring–Fall 2026

Contents

1 Your Role

Your Scope Only

You own: Three-track CV pipeline — Worker Detection, PPE (helmet/vest), Hazard (fire/smoke/spill) — plus compliance engine, evidence capture, and a stable **handoff dict** for Member A. All work in `edge/inference/` and `edge/compliance_engine/`.

Not Your Responsibility

You must NOT: Write MQTT code, configure Mosquitto, deploy to Jetson, touch `transport/`, `services/`, `frontend/`, or `infra/`. Member A deploys your merged PRs and publishes events.

1.1 Supervisor-approved classes (6 minimum)

Track	Classes
Worker Detection	`worker`
PPE Detection	`helmet`, `vest`
Hazard Detection	`fire`, `smoke`, `spill`
Optional stretch	`gloves`, `boots` — only if easy
Out of scope	`goggles`, `forklift`

See `finalization/DETECTION_CLASS_SCOPE.md`.

2 Sprint 1 (Weeks 1–4): Dataset & Handoff Spec

Your focus — Data pipeline and integration boundary

Week 1–2

- Python venv; install OpenCV, Ultralytics, PyTorch; verify GPU/CUDA
- Clone repo; branch `feature/b-edge-readme`; first PR
- Source PPE/hazard datasets (Roboflow, Kaggle, custom capture)
- Set up LabelImg / Roboflow / CVAT; YOLO label format
- Create `edge/inference/README.md` — 6 classes, three tracks diagram

Week 3–4

- Annotate first 200+ images (all 6 classes represented)
- Draft **CV handoff interface** in `edge/README.md`:

```
get_compliance_event(frame) -> dict
# event, zone, worker_id, compliance_score,
# confidence, detections[], image_path
```

- Run YOLO pretrained inference demo; save annotated sample outputs
- Sign prerequisites checklist

Your Sprint 1 done when: Handoff spec reviewed by Member A; 200+ labels committed; pretrained demo runs locally.

Collaboration Touchpoint

Week 4 with Member A: Align handoff dict fields with `contracts/events.py` (**M0**). You do not attend MQTT setup — only review contract mapping document.

3 Sprint 2 (Weeks 5–8): Training & Compliance Prototype

Your focus — YOLO training + IoU + compliance logic

Week 5–6

- Expand dataset to 500+ images; `data.yaml` for Ultralytics
- Train YOLOv8n; monitor mAP on val set
- Implement IoU function: associate helmet/vest boxes to nearest worker
- Unit tests for IoU and debounce logic

Week 7–8

- Compliance engine prototype in `edge/compliance_engine/`
- Per-zone rules: helmet + vest required (config file, not DB — A owns DB)
- Rolling 5-frame compliance score
- Evidence: `cv2.imwrite` on violation; return `image_path` in handoff dict
- Violation debounce (60s window per worker/event)

Your Sprint 2 done when: Local demo: webcam → inference → handoff dict printed to console; mAP trending ≥ 0.60 on val.

Collaboration Touchpoint

Week 8: Demo handoff JSON file to Member A for their test harness (**M1 prep**). Answer questions in integration session — no Jetson required yet.

4 Sprint 3 (Weeks 9–12): Model Finalization & PR Merge

Your focus — Hit mAP target; export model; merge PRs

Week 9–10

- Dataset 800–1000+ images; fix weak classes (fire/smoke often need more data)
- Retrain; target mAP@0.5 ≥ 0.70 on held-out test set
- Confusion matrix + per-class report in `edge/inference/EVAL.md`
- Hazard track: fire/smoke/spill events independent of worker IoU

Week 11–12

- Export best weights `.pt` + ONNX for Member A's TensorRT deploy
- Finalize `edge/compliance_engine/`; open PRs; address A's review comments
- `requirements.txt` + setup instructions for Jetson inference deps
- Optional stretch: gloves/boots only if mAP ≥ 0.70 without hurting core 6 classes

Your Sprint 3 done when: PRs merged; **M2** — Member A confirms handoff harness works with your latest dict output.

Collaboration Touchpoint

Week 12: Deliver model package (weights + ONNX + `EVAL.md`) via agreed storage (Drive/Git LFS). Do not SSH to Jetson.

5 Sprint 4 (Weeks 13–16): Jetson Tuning Support

Your focus — Support deploy feedback; no deploy yourself

Week 13–14

- Fix bugs filed by Member A from Jetson deploy (Jira tickets assigned to you)
- Tune confidence thresholds if false positives on edge camera
- Re-export ONNX if architecture changes
- Document known limitations in `edge/README.md`

Week 15–16

- On-site or remote support during Member A's integration sessions
- Verify: worker + helmet + vest association stable at Jetson FPS (>15 target)
- Hazard detection spot-check near welding zone test footage
- Prepare CV slides for mid/final presentation (your sections only)

Your Sprint 4 done when: **M3** — Member A confirms Jetson CV + your handoff works; your open Jira blockers = 0.

Collaboration Touchpoint

Week 16: Attend integration session; explain compliance logic live if supervisor asks. Member A runs hardware.

6 Sprint 5 (Weeks 17–20): Polish, Docs & Viva

Your focus — CV documentation and viva readiness

Week 17–18

- Final threshold tuning from demo rehearsal feedback
- Write FYP report sections: dataset, training, evaluation, compliance engine
- Architecture diagram: three tracks → handoff → (Member A's MQTT)
- Prepare 5-minute CV demo script segment for joint presentation

Week 19–20

- Benchmark note: FPS + mAP on Jetson (Member A runs; you document numbers)
- Viva prep: explain IoU, debounce, mAP, why 6 classes, handoff boundary
- Review full system architecture (high level) — not only your module
- Final sprint log; close all `member-b` Jira tickets

Your Sprint 5 done when: M4 — joint demo successful; CV report sections submitted; you can explain every line in your merged PRs.

Collaboration Touchpoint

Week 20: Final demo — you present CV/detection slides; Member A operates platform. Q&A split by domain.

7 Your Master Checklist

- ☐ 1000+ annotated images (6 classes)
- ☐ YOLOv8n mAP@0.5 ≥ 0.70
- ☐ Worker / PPE / Hazard three-track pipeline
- ☐ IoU association (helmet + vest → worker)
- ☐ Compliance engine + debounce + 5-frame score
- ☐ Evidence capture with `image_path`
- ☐ Handoff dict documented and tested by Member A
- ☐ ONNX + weights delivered; PRs merged

- ☐ Zero MQTT / Jetson deploy commits under your name
- ☐ Weekly sprint logs + Jira tickets closed per sprint

8 Handoff Dict Reference (Your Output Boundary)


```
{
  "event": "helmet_missing | vest_missing | fire | ...",
  "zone": "welding_bay",
  "worker_id": 7,
  "compliance_score": 78,
  "confidence": 0.92,
  "detections": [{"class": "helmet", "bbox": [...], "confidence": 0.94}],
  "image_path": "evidence/session_id/frame_001.jpg"
}
```


Member A adds `type`, `priority`, `timestamp`, `session_id` and MQTT publish.